

I used one harness for multiple model APIs

- I used model APIs to choose actions and built a harness to execute those actions in Slack.
- The harness owns the starting data, browser, result checks, and recording, so changing models does not require rebuilding the test.

MODEL INPUT



What does it see?

Compare screenshots, page controls, and structured data.

ACTION EXECUTION



How does it act?

Turn the response into a browser action or an API operation.

RESULT CHECK

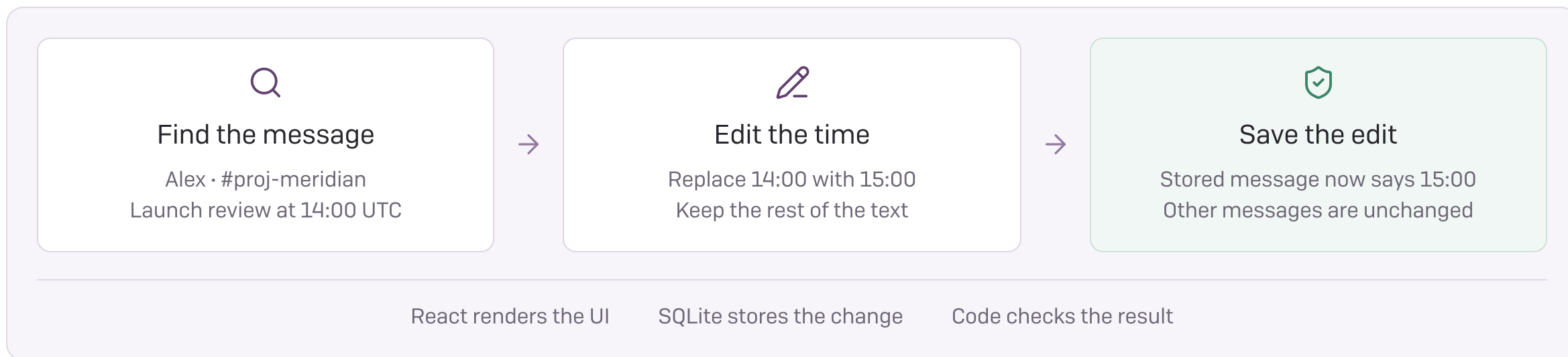


Did the change happen?

Check saved application data, not the model's claim.

I started with a task that needs a real saved change

- The agent must find Alex's launch-review message and change its time from 14:00 to 15:00 UTC.
- Typing the new time is not enough: it must save the edit without changing other messages.



- A static mockup cannot test persistence. A full Slack backend adds services these tasks do not need.
- Fictional data · Task abbreviated here · [Try the workspace without a key](#) · [Prior-work review](#).

The task set covers 18 workflows

Six focused tasks

Update a topic

Channel → edit topic → save

Reply in a thread

Find request → open thread → reply

Edit a message

Find own message → edit → save

Triage an incident

Find active incident → react → pin

Delete a draft

Find own draft → delete → confirm

Send a handoff

Read current details → compose DM

Coordinate work

Coordinate a release *

Read sources → edit → reply → topic → DM

Close out an incident

Read resolution → replace pin → reply → topic

Refresh saved work

Find current sources → replace saved items → DM

Record a design decision

Read approval → react → pin → save → details

Repair an outdated handoff

Read correction → edit DM → react → save

Complete a QA sign-off

Read test results → reply → fix reaction → save

Combine several changes

Publish a release update

Read source → format update → pin → delete draft

Brief the on-call engineer

Read rotation → DM primary → save → details

Correct a threaded estimate

Read review → edit own reply → react → reply

Refresh the launch board

Find plan → replace pin → save → topic

Hand off an approved design *

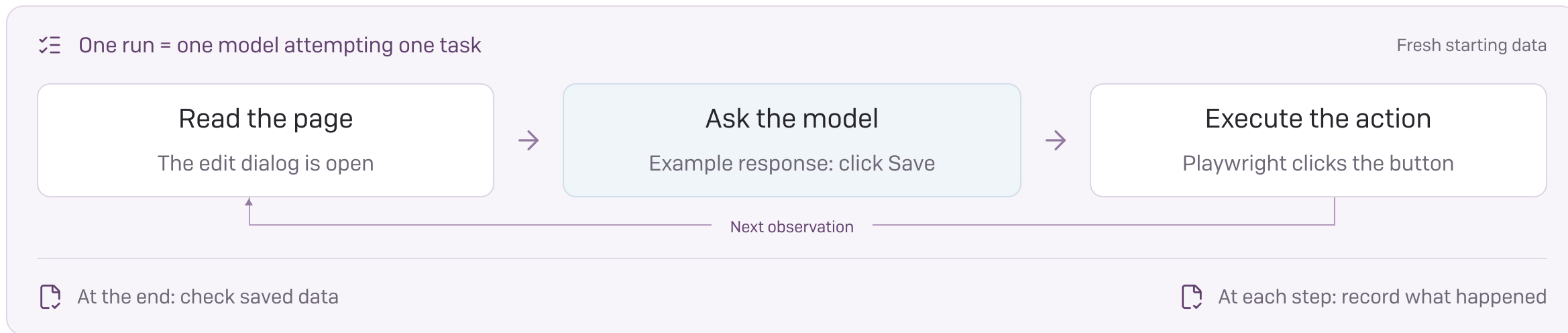
Read decision → react → DM → reply → save

Prepare a release retrospective

Read three sources → summarize → pin → save → DM

- These diagrams summarize the intended workflows. Exact instructions and preservation checks remain in each task.
- * Two original tasks contain wording problems. Their results remain unchanged and are discussed later.

The model proposes an action; the harness executes it



- The harness accepts only supported actions, so a model response cannot execute arbitrary code.
 - After each action, it reads the updated page before asking the model what to do next.
 - The run stops when the model finishes or reaches an action, time, or spending limit.
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- The example shows browser control. The API path executes application operations instead.

The model can use four interfaces to the same app

Interface	Observation → action	Interaction
 Pixels	Screenshot → mouse and keyboard	Browser
 Accessibility	Named page controls → click or fill	Browser
 Page JSON	Structured page controls → click or fill	Browser
 Actor API	Workspace data → direct changes	Tool use

- The interface determines what the model observes and which actions it can request.
- The actor API uses the same application rules but does not operate the browser.
- All four paths change workspace data that the same grader can check.

- Page JSON describes page controls, not the underlying database. Actor API access exposes broader structured data.
- The harness uses a shared JSON action protocol, not each provider's native computer-use protocol.

I grade saved data, not visible text or a completion message

- The grader is code that checks the stored messages, topics, and other workspace data.
- For the editing task, the original message must contain the requested text and all unrelated data must stay unchanged.



Required saved message

“Launch review is at 15:00 UTC. Please bring the final checklist.”



Right change

Exact text in Alex's original message



No extra changes

No new message or unrelated edit

Both pass

Reward = 1

- The grader checks the final state. It does not penalize a wrong action that the agent fully undoes.
- Task definitions and result checks.

I tested correct and nearly correct outcomes separately

CAN THE UI DO IT?

Run a scripted solution

Search → edit → Save → inspect the stored message

The requested change must persist.

CAN THE GRADER CATCH A NEAR MISS?

Alter a known correct result

Wrong destination, wrong author, or one extra change

The altered result must fail.

CAN I EXPLAIN A MODEL'S RESULT?

Reopen the recorded run

Read its actions and check its saved data again

The recorded grade must agree.

- These checks separate application bugs, grading bugs, and agent mistakes without paying for more model calls.
- A scripted solution still knows my intended answer, so it cannot prove that a new reader understands the task.

• [Software verification](#) • The longer-workflow grader tests cover 84 valid and 2,583 invalid states across seven seeds.

Two agents must not share mutable workspace data

- If agent A edits a message, agent B must still see its own unchanged copy.
- I give each run a separate SQLite file and a fresh browser context, which is an independent browser session.

Shared application code + same starting fixture

RUN A

Browser context A → database A

Agent saves its edit

Launch review at 15:00

RUN B

Browser context B → database B

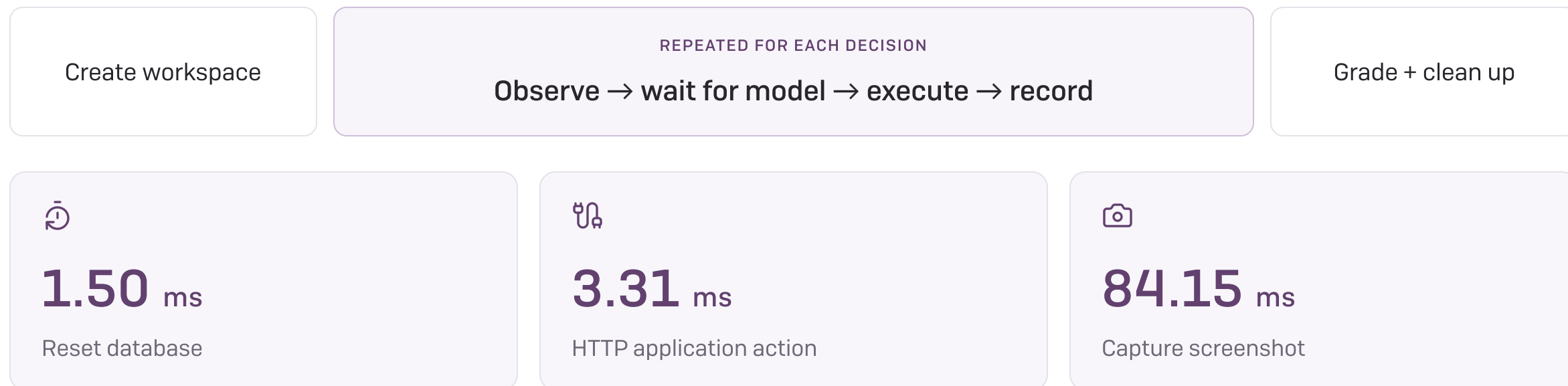
No action in this run

Launch review stays at 14:00

A session token selects one database. The agent is not given another run's token.

- Tested: write to A, check that B is unchanged; close A, check that B still works.
- The actor and control servers share a process. Separate data is not isolation from host crashes or malicious code.

Run time comes from repeated steps, not just reset



- Every extra decision adds another model request and another observation.
- A faster database reset helps between runs, but does not remove the cost of those repeated steps.

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- Measured locally on M5 Max / Node 24. Component medians exclude model calls and cannot be added into an end-to-end estimate.
 - Samples and setup: 100 stored sessions; screenshots from 10 sequential browser contexts, not 100 concurrent agents.

Small databases do not mean cheap browser runs

WORKSPACE DATA

One SQLite file per run

Simple creation, reset, and deletion

Measured: 48 KiB per fixture after one post

BROWSER EXECUTION

Fresh browser per experiment run

No browser state carried into the next run

Cost: startup time and browser memory

EVIDENCE

Record actions and workspace states

Review failures without new model calls

Cost: growing files and disk writes

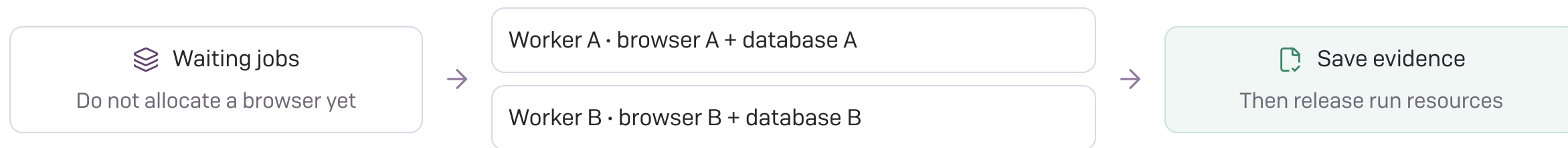
- I share application code, but keep each run's data separate.
- The storage test did not measure full browser memory, so it cannot tell us how many agents fit on a machine.

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- The experiment runner creates and closes an environment per run. Reusing a browser across fresh contexts is a future optimization.
 - API execution can omit the browser when visual recording is disabled. The reported interface study kept observer recording enabled.

I would scale with a fixed number of run workers

- The sequential collections do not demonstrate multi-agent throughput.
- A queue would admit work only when a worker has capacity, instead of launching a browser for every incoming request.

PROPOSED, NOT IMPLEMENTED OR LOAD-TESTED



More workers can finish more runs at once. They do not make an individual model request faster.

- Test 1, 4, 8, and 16 workers: successful runs/hour, full process-tree memory, provider limits, and cleanup after a killed worker.
- Use separate processes or containers for stronger containment. Browser pooling trades lower startup cost for shared failure risk.

The reusable harness made model comparisons possible

- Once the harness worked, I could change the model without rebuilding the test.
- I expanded the assignment into a benchmark prototype with two separate comparisons.
- The first compares models using accessibility controls, and the second compares four interfaces.

Original assignment



Working environment

Focused Slack workflows
Known starting state



Repeatable harness

Reset · act · check · record
Select a model



Expanded scope



Benchmark prototype

306 model-comparison runs
96 interface-comparison runs

The next slides cover the test conditions, results, and limitations of each comparison.

Comparison 1: 17 models, using accessibility controls

$$\begin{array}{ccccccc} \text{📁} & & \text{☑️} & & \text{🛡️} & & \\ 17 & \times & 18 & \times & 1 & = & 306 \\ \text{model routes} & & \text{tasks} & & \text{attempt per pair} & & \text{recorded attempts} \end{array}$$

- Each model receives named page controls and returns one JSON action per request.
- All 306 runs use the same starting data, recent-four history, and no site guide.
- This tests model performance under one interface, not the best interface for each model.

40 actions 180 s per run \$5 estimated allowance per run

- Seed 1042 • 4,096 output tokens per request • identical graders • provider-default reasoning.
- I kept 105 earlier runs and added 201 new runs without replacing failures.
- [Node 24 setup and API key instructions](#) • [Exact routes, limits, and run history](#). Provider routing and collection time were not controlled.

Results for 17 models

1 per task/model · Collection complete

306 / 306 attempted · 152 passed · 61 incomplete · 93 blocked

\$189.3110

Usage estimate

\$3.7794

Unresolved · 35 calls

\$193.0904

Total allowance · 306 trials

Model ↕	Trials ↕	Passed ↕	Incomplete ↕	Blocked ↕	Median time ↕	Usage estimate ↕	Unresolved ↕	Total allowance ↕
 GPT-6 Astra	18 / 18	14 / 18	1	3	27.1s	\$35.9623	\$0.0000	\$35.9623
 GPT-6.1 Sol	18 / 18	14 / 18	4	0	28.5s	\$8.4352	\$0.0000	\$8.4352
 GPT-6 Luna	18 / 18	9 / 18	8	1	39.4s	\$0.5394	\$0.0092	\$0.5486
 GPT-OSS 120B	18 / 18	5 / 18	2	11	180.1s	\$0.6682	\$0.1358	\$0.8040
 Claude Fable 5.1	18 / 18	13 / 18	1	4	62.8s	\$48.4714	\$0.0000	\$48.4714
 Claude Opus 5.5	18 / 18	13 / 18	4	1	38.3s	\$28.6987	\$0.3741	\$29.0728
 Claude Sonnet 5.5	18 / 18	7 / 18	4	7	39.9s	\$13.9323	\$0.3304	\$14.2627
 Claude Haiku 4.5	18 / 18	6 / 18	3	9	28.3s	\$3.7921	\$0.0000	\$3.7921
 Grok 4.7	18 / 18	10 / 18	6	2	34.3s	\$11.9101	\$0.3809	\$12.2910
 Qwen 3.8 Max	18 / 18	7 / 18	1	10	125.1s	\$6.9999	\$1.0639	\$8.0638
 DeepSeek V4 Pro	18 / 18	8 / 18	1	9	106.8s	\$4.6816	\$0.2547	\$4.9363
 DeepSeek V4.1 Flash	18 / 18	7 / 18	3	8	65.9s	\$1.3825	\$0.0541	\$1.4365
 GLM 5.3	18 / 18	7 / 18	0	11	125.3s	\$3.6307	\$0.3453	\$3.9760
 GLM 5.3 Flash	18 / 18	8 / 18	0	10	46.2s	\$0.4444	\$0.0000	\$0.4444
 Kimi K3	18 / 18	11 / 18	3	4	50.0s	\$14.0051	\$0.7958	\$14.8009
 MiniMax M3	18 / 18	5 / 18	10	3	25.9s	\$1.4698	\$0.0353	\$1.5052
 Nemotron 3 Ultra	18 / 18	8 / 18	10	0	50.1s	\$4.2874	\$0.0000	\$4.2874

Passed Incomplete Blocked Not run

- 18 public tasks. 1 run per task and model. All use accessibility controls. Median time includes blocked runs.
- Total allowance adds the usage estimate to money reserved for missing usage. These are estimates, not invoices. Including earlier tests: \$194.1058 of the \$300 budget. [Full costs CSV](#) · [JSON](#)
- 105 earlier runs are included. No retries. One run per task cannot establish a reliable model ranking. Two tasks have unclear instructions. [Read the failure analysis.](#)
- [Test plan](#) · [Recorded runs](#) · [Download all trials \(CSV\)](#)

A correct grader can still test an unclear instruction

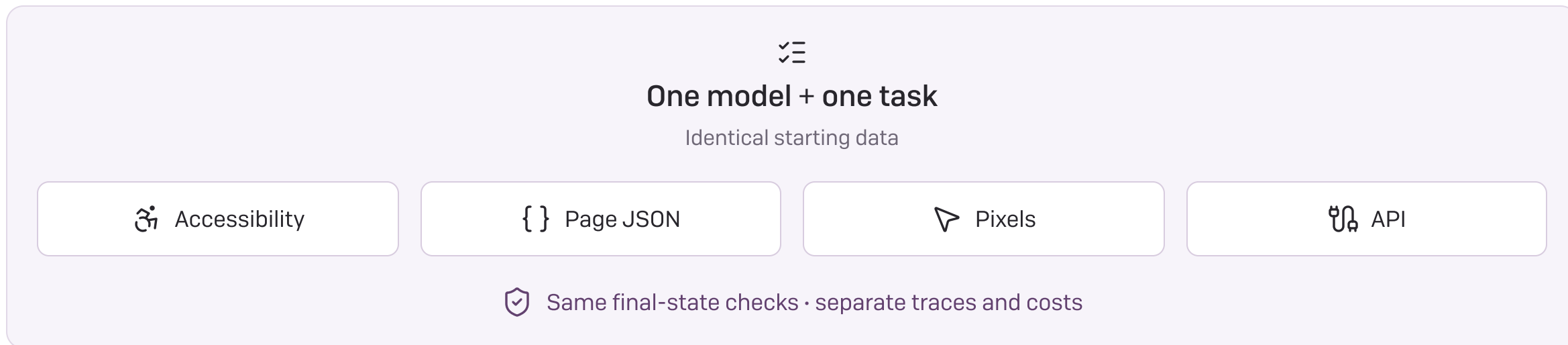
- One task asks the agent to send the approved design to a teammate.
- The workspace names the design Willow, but my task quotes a message containing DESIGN without explaining the substitution.

INSTRUCTION EXCERPT	DM its named handoff recipient “Implement DESIGN navigation; accessibility approved.”
SIX COMPLETED RUNS SENT	“Implement DESIGN navigation; accessibility approved.”
THE GRADER REQUIRED	“Implement Willow navigation; accessibility approved.”

Missing check: could an independent reader get the same answer from this instruction?

- This explains a problem in six traces, not all 17 failures. A second task had an unclear source location.
- [Recorded actions and failure analysis](#). Keep the recorded grades. Fix the instruction in a new task version before collecting new results.

Comparison 2: the same tasks through four interfaces



- I ran four models on six tasks through four interfaces, giving 96 fresh attempts.
- Each attempt starts fresh, and the interface order is randomized within each model and task.
- This is a separate study with a \$25 allowance and no retries.

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- Sol, Sonnet, Qwen and Grok • topic, thread reply, message edit, incident closeout, saved cleanup, decision record.
 - Seed 2042 • low reasoning • recent-four history • no guide • 40 actions • 180 seconds • \$1 ceiling per cell.

Interface comparison: 96 recorded attempts

Passed / attempted · median elapsed time in seconds · six tasks per cell

Model ↕	Accessibility ↕	Page JSON ↕	Pixels ↕	Actor API ↕	Allowance ↕
 GPT-6.1 Sol	<u>6 / 6</u> 0 incomplete · 0 blocked All 28.4 s Passed 28.4 s	<u>5 / 6</u> 0 incomplete · 1 blocked All 33.2 s Passed 50.3 s	<u>1 / 6</u> 0 incomplete · 5 blocked All 107.4 s Passed 76.7 s	<u>6 / 6</u> 0 incomplete · 0 blocked All 10.0 s Passed 10.0 s	\$3.9430
 Claude Sonnet 5.5	<u>4 / 6</u> 0 incomplete · 2 blocked All 26.3 s Passed 10.2 s	<u>4 / 6</u> 1 incomplete · 1 blocked All 26.5 s Passed 10.8 s	<u>2 / 6</u> 1 incomplete · 3 blocked All 23.7 s Passed 19.2 s	<u>5 / 6</u> 1 incomplete · 0 blocked All 9.9 s Passed 7.8 s	\$7.2784
 Qwen 3.8 Max	<u>5 / 6</u> 1 incomplete · 0 blocked All 26.7 s Passed 14.7 s	<u>4 / 6</u> 1 incomplete · 1 blocked All 23.9 s Passed 36.6 s	<u>0 / 6</u> 0 incomplete · 6 blocked All 55.2 s Passed N/A	<u>5 / 6</u> 0 incomplete · 1 blocked All 11.9 s Passed 6.3 s	\$5.4829
 Grok 4.7	<u>3 / 6</u> 1 incomplete · 2 blocked All 20.8 s Passed 7.8 s	<u>3 / 6</u> 3 incomplete · 0 blocked All 17.2 s Passed 7.8 s	<u>0 / 6</u> 1 incomplete · 5 blocked All 40.5 s Passed N/A	<u>5 / 6</u> 1 incomplete · 0 blocked All 9.0 s Passed 5.2 s	\$6.0573

[All traces and replays ↗](#) [Costs CSV ↓](#) [Paired results and timing ↗](#)

- All-attempt medians include early stops, while passed-run medians use only successful tasks.
- Passed tasks differ across cells, so these medians do not establish a speed ranking.

- Verified: 58 passed · 11 incomplete · 27 blocked. Usage estimate \$20.7180 + unresolved \$2.0435 = \$22.7616 of \$25.
- N/A means no passed task. Time includes model calls and browser work, not just model latency. Click a pass count for traces and replays.

The API result does not show better computer use

EXAMPLE: API TOPIC CHANGE



Set the topic directly

One application operation
No dialog to navigate

SAME CHANGE IN THE BROWSER



Open → fill → Save

Several UI actions
Each requires another decision

PIXEL RESULTS



19 / 24 stopped early

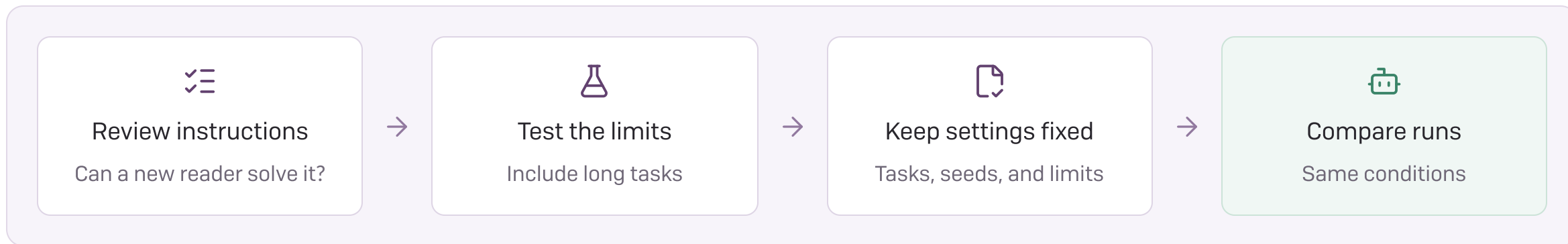
14 connection failures
3 output limits • 2 timeouts

- The same final check does not make the routes equally difficult: API access skips browser interaction.
- A pixel failure can reflect image delivery, keyboard behavior, or a service interruption, not just a wrong decision.
- One attempt per condition cannot establish a reliable winner.

- 17 shared passes for API/accessibility. 24 API/accessibility pairs: 17 both passed • 4 API only • 1 accessibility only • 2 neither passed. 5 pairs include a blocked attempt.
- On 17 shared passes, median accessibility-minus-API differences were +2 actions, +5.462 seconds, and +\$0.13008. These are selected successful pairs, not a general speed estimate.
- Pixels used low-detail images on macOS. Time includes shared-host activity. [Definitions, costs, and all matched pairs.](#)

The next version needs validated tasks and repeated runs

- Ask someone outside the project to solve each task from its instructions.
- Compare their interpretation with the expected result.
- Set run limits with both short and long tasks.
- Use new task variants and repeat runs to measure reliability.



The current results describe these attempts. Repeated runs are needed to estimate reliability.

- [Detailed engineering review](#) · [Remaining validation work](#). These are next steps, not completed results.

Demo and design questions



01 · TRY

Use the workspace

Search, open a thread, edit



02 · COMPARE

Choose a model and task

All 306 attempts and costs



03 · REVIEW

Inspect the run

Actions, changes, result checks

- Which workflows should the next version test?
- Should the score include wrong actions that the agent later undoes?
- How many agents must run at once, and what should each run cost?

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- Relay can reset, execute actions, and return a reward. No reinforcement learning training has been run.
 - [Code and setup](#) · [Requirements and remaining work](#).